MATH 152
Mrs. Bonny Tighe

EXAM II

50 points
7.7-11.7
PART I

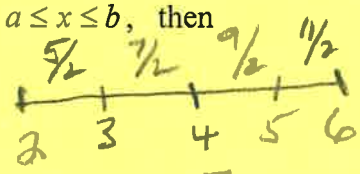
NAME Solutions

TA_____ Wed 11/3/2021

1. (10 points) Use (a) the Midpoint Rule (b) the Trapezoidal Rule and (c) Simpson's Rule to approximate the given integral with the specified value of n. Find the error bounds for the Midpoint rule. Suppose $|f''(x)| \le K$ *for* $a \le x \le b$, then

$$|E_M| \le \frac{K(b-a)^3}{24n^2} \qquad \int_2^6 \ln x\, dx, \quad n = 4$$

$\Delta x = \frac{4}{4} = 1$



$M_4 = 1\left[\ln(5/2) + \ln(7/2) + \ln(9/2) + \ln(11/2)\right]$

$T_4 = \frac{1}{2}\left[\ln 2 + 2\ln 3 + 2\ln 4 + 2\ln 5 + \ln 6\right]$

$S_4 = \frac{1}{3}\left[\ln 2 + 4\ln 3 + 2\ln 4 + 4\ln 5 + \ln 6\right]$

$f'(x) = \frac{1}{x} = x^{-1}$

$f''(x) = -x^{-2}$

$\left|-\frac{1}{x^2}\right|$

largest when $x = 2$

So $K = \frac{1}{4}$

$|E_M| \le \frac{1/4\,(6-2)^3}{24(4)^2}$

$|E_M| \le \frac{\frac{1}{4}(4)^3}{24(4)^2} = \frac{1}{24}$

How large should we take n in order to guarantee that the Midpoint approximation for $\int_2^6 \ln x\, dx$ is accurate to within 0.001?

$|E_M| \le \frac{\frac{1}{4}(4)^3}{24n^2} \le \frac{1}{10^3}$

$\le \frac{16}{24n^2} \le \frac{1}{10^3}$

$\frac{2}{3n^2} \le \frac{1}{10^3}$

$\frac{2000}{3} \le n^2$

$\sqrt{\frac{2000}{3}} \le n$

The next larger integer

2. (15 points) Multiple Choice: Circle the correct letter

a) Find a sequence for $1, \frac{1\cdot 5}{1}, \frac{1\cdot 5\cdot 9}{1\cdot 2}, \frac{1\cdot 5\cdot 9\cdot 13}{1\cdot 2\cdot 3}, \ldots$ is

a) $a_n = \frac{4n-3}{n!}$

b) $a_n = \frac{(1\cdot 5\cdot 9\cdots(4n-3))}{(n-1)!}$

c) $a_n = \frac{(1\cdot 5\cdot 9\cdots(4n-3))}{n!}$

d) $a_n = \frac{4n-1}{(n-1)!}$

e) None of the above

b) $\sum_{n=1}^{\infty} \frac{(-3)^{n+1}}{2^{2n}}$ equals

$\sum_{n=1}^{\infty} \frac{(-3)^2(-3)^{n-1}}{4\cdot 4^{n-1}}$ $r = |-3/4|$ convergent

$\sum_{n=1}^{\infty} \frac{9}{4}\left(\frac{-3}{4}\right)^{n-1}$ $S = \frac{9/4}{1-(-3/4)}$

$S = \frac{9}{4+3}$

a) $\frac{-12}{7}$

b) $\frac{9}{7}$

c) 12

d) $\frac{-9}{7}$

e) None of the above

c) Which of the following sequences is not convergent?

a) $a_n = \frac{5^n+2}{3^n}$

b) $a_n = \frac{4n-1}{2n+1}$

c) $a_n = \frac{3}{n!}$

d) $a_n = \ln(n+2) - \ln(n+3)$

e) They are all convergent

d) Which test would you use to determine the absolute convergence of the series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n^3+4n+2}}$$

a) Alternating Series Test
b) Integral Test
c) Limit Comparison Test
d) Ratio Test
e) None of the above tests

e) Which test would you use to determine the convergence of the series $\sum_{n=0}^{\infty} \frac{(-1)^n}{e^n}$

a) Limit Comparison Test
b) Root Test
c) Integral Test
d) Ratio Test
e) All of the above

3. (5 points) Circle True or False

a) **True** **False** The first test you try when determining the convergence of a series is the limit comparison test.

b) **True** **False** Given $a_n = n^2 + 2n + 1$, then $a_{n+1} = n^2 + 4n + 4$. $(n+1)^2 + 2(n+1) + 1$ $n^2 + 2n + 1 + 2n + 2 + 1$

c) **True** **False** To determine the convergence of $\sum_{n=0}^{\infty} \frac{n-1}{n^2+n^3+1}$ I could use the Integral Test.

d) **True** **False** If $\lim_{n\to\infty} a_n = 0$, then the series $\sum_{n=0}^{\infty} a_n$ is convergent.

e) **True** **False** If $\lim_{n\to\infty} a_n = 0$, then the sequence $\{a_n\}$ is convergent.

4. (10 points) Circle True or False

True **False** a) The formula for the Trapezoid rule for approximation of a definite integral with four subdivisions is $\Delta x\left[f(x_0)+2f(x_1)+2f(x_2)+2f(x_3)+f(x_4)\right]$.

True **False** b) We would compare $\int_1^\infty \frac{x}{\sqrt{x^6+2x-3}}dx$ to $\int_1^\infty \frac{1}{x^2}dx$ to determine convergence.

True **False** c) The arc length of the function $y=\sin x$ on the interval $0 \le y \le 1$ is $\int_0^1 \sqrt{1+\left(\frac{1}{1-y^2}\right)}dy$.

$x = \sin^{-1} y$

$dy/dx = \frac{1}{\sqrt{1-y^2}}$

True **False** d) $\sum_{n=2}^{\infty} \frac{(-1)^n n}{\ln(n)}$ is conditionally convergent. Divergent

True **False** e) The Surface Area for the function $y=e^{2x}$ on the interval $0 \le x \le 1$ is

$2\pi\int_0^1 e^{2x}\sqrt{1+4e^{2x^2}}dx$

forgot to add axis

5. (15 points) Multiple Choice – (3 points each) Circle the correct letter

a) When using the comparison test for the improper integral $\int_2^\infty \frac{x}{\sqrt{x^5-2x^3-3}}dx$ the inequality will be

a) $\frac{1}{\sqrt{x^5}} \le \frac{x}{\sqrt{x^5-2x^3-3}}$

b) $\frac{1}{\sqrt{x^5}} \ge \frac{x}{\sqrt{x^5-2x^3-3}}$

c) $\frac{1}{\sqrt{x^3}} \le \frac{x}{\sqrt{x^5-2x^3-3}}$

d) $\frac{1}{\sqrt{x^3}} \ge \frac{x}{\sqrt{x^5-2x^3-3}}$

e) None of the above

$x^5-2x^2-3 \ge x^5$

$-2x^2-3 \ge 0$

$r = x$

b) The surface area for the function $x = e^y$ when rotated about the y-axis on the interval $2 \le y \le 3$ is:

dy

$dx/dy = e^y$

a) $2\pi \int_2^3 y\sqrt{1+e^{2y}}\,dy$

b) $2\pi \int_2^3 e^y \sqrt{1+e^{2y}}\,dy$

c) $2\pi \int_2^3 x\sqrt{1+\left(\frac{1}{x^2}\right)}\,dx$

d) $2\pi \int_2^3 \ln x\sqrt{1+\left(\frac{1}{x^2}\right)}\,dx$

e) None of the above

c) The K for the error bound for the midpoint formula for approximating $\int_1^4 \frac{1}{\sqrt{x}}\,dx$ with 4 subdivisions is:

$f(x) = x^{-1/2}$

$f'(x) = -\frac{1}{2}x^{-3/2}$

$f''(x) = \frac{3}{4}x^{-5/2}$

$= \frac{3}{4x^{5/2}}$

largest at $x = 1$

a) $K = -\frac{3}{4}$

b) $K = \frac{3}{128}$

c) $K = \frac{3}{4}$

d) $K = \frac{1}{2}$

e) None of the above

d) Which of the following is **not** an improper integral?

a) $\int_{-2}^{5} \frac{1}{\sqrt{x+2}}\,dx$

b) $\int_1^3 \frac{x}{x^2+3x-4}\,dx$

c) $\int_1^3 \frac{x}{x^2-3x+4}\,dx$

d) $\int_0^1 \frac{1}{\ln x}\,dx$

e) They are all improper integrals

e) Represent $0.\overline{746}$ as a series

$\frac{746}{10^3} + \frac{746}{10^3}\left(\frac{1}{10^3}\right) +$

a) $\sum_{n=0}^{\infty} \left(\frac{746}{1000}\right)^n$

b) $\sum_{n=0}^{\infty} \frac{746}{1000}\left(\frac{1}{10}\right)^n$

c) $\sum_{n=0}^{\infty} 746\left(\frac{1}{1000}\right)^n$

d) $\sum_{n=0}^{\infty} \frac{746}{1000}\left(\frac{1}{1000}\right)^n$

e) None of the above

MATH 152 **EXAM II** NAME Solutions

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7.7-11.7 TA_____ Wed 11/3/2021

PART II

1. (10 points) Complete the following tests

GEOMETRIC SERIES $\sum_{n=0}^{\infty} a_n (r)^n$ is divergent if $|r| \geq 1$

THE COMPARISON TEST Suppose $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are series with positive terms.

i) If $\sum_{n=1}^{\infty} b_n$ is convergent and $a_n \leq b_n$ for all n, then $\sum_{n=1}^{\infty} a_n$ is also convergent

ii) If $\sum_{n=1}^{\infty} b_n$ is divergent and $a_n \geq b_n$ for all n, then $\sum_{n=1}^{\infty} a_n$ is also divergent

THE LIMIT COMPARISON TEST

Suppose that $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are series with positive terms. If $\lim_{n\to\infty} \frac{a_n}{b_n} = c$ where c is > 0 and finite, then either both series converge or both diverge.

Name the test: Test for Divergence If $\lim_{n\to\infty} a_n \neq 0$, then $\sum_{n=1}^{\infty} a_n$ is divergent

THE INTEGRAL TEST Suppose f is a positive, decreasing and continuous function on $[1, \infty)$ and let $a_n = f(n)$ then the series $\sum_{n=1}^{\infty} a_n$ is convergent if and only if $\int_0^{\infty} f(n)\,dn$ is convergent.

P-SERIES $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is convergent if $p > 1$

RATIO TEST

If $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = L$

$L < 1$ then $\sum a_n$ is absolutely convergent

$L > 1$ then $\sum a_n$ is divergent

$L = 1$ then the test is inconclusive

ROOT TEST

If $\lim_{n\to\infty} \sqrt[n]{|a_n|} = L$ and

$L < 1$ then $\sum a_n$ is absolutely convergent

$L > 1$ then $\sum a_n$ is divergent

$L = 1$ then the root test is inconclusive

THE ALTERNATING SERIES TEST

If i) if $b_{n+1} \leq b_n$ for all n and ii) $\lim_{n\to\infty} b_n \neq 0$ Then $\sum_{n=1}^{\infty} (-1)^{n-1} b_n$ is Divergent

REMAINDER ESTIMATE FOR THE INTEGRAL TEST

Suppose $a_n = f(n)$ where f is a continuous, positive, decreasing function for $x \geq n$, and $\sum a_n$ is convergent. If $R_n = S - S_n$, then $\int_{n+1}^{\infty} f(n)\,dn \leq R_n \leq \int_n^{\infty} f(n)\,dn$

2. (10 points). Use the **Integral test** to determine if the series is divergent or convergent.

$$\sum_{n=1}^{\infty} \frac{e^n}{e^{2n}+1}$$

$f(n) = \frac{e^n}{e^{2n}+1}$ which is continuous, positive and decreasing for $n \geq 1$

$$\lim_{t\to\infty} \int_1^t \frac{e^n}{e^{2n}+1}\,dn = \lim_{t\to\infty} \tan^{-1}(e^n)\Big]_1^t$$

$$\lim_{t\to\infty} \left[\tan^{-1}(e^t) - \tan^{-1}(e^1)\right] = \pi/2 - \tan^{-1}(e)$$ which is finite so is convergent

$\sum_{n=1}^{\infty} \frac{e^n}{e^{2n}+1}$ is also convergent

3. (10 points) Find the length of the arc of the curve $x = \frac{y^4}{8} + \frac{1}{4y^2}$ $1 \leq y \leq 2$

$$L = \int_1^2 \sqrt{1 + (dx/dy)^2}\,dy$$

$$L = \int_1^2 \sqrt{\tfrac{1}{4}y^6 + \tfrac{1}{2} + \tfrac{1}{4y^6}}\,dy$$

$$L = \int_1^2 \sqrt{\left(\tfrac{1}{2}y^3 + \tfrac{1}{2y^3}\right)^2}\,dy$$

$$L = \int_1^2 \left(\tfrac{1}{2}y^3 + \tfrac{1}{2}y^{-3}\right)dy$$

$$L = \tfrac{1}{8}y^4 - \tfrac{1}{4}y^{-2}\Big]_1^2$$

$$\left(\tfrac{1}{8}(2^4) - \tfrac{1}{4(4)}\right) - \left(\tfrac{1}{8} - \tfrac{1}{4}\right)$$

$$2 - \tfrac{1}{16} - (-1/8)$$

$$2 - \tfrac{1}{16} + \tfrac{1}{8} = 2 + 1/16$$

$$x = \tfrac{1}{8}y^4 + \tfrac{1}{4}y^{-2}$$

$$dx/dy = \tfrac{1}{2}y^3 - \tfrac{1}{2}y^{-3}$$

$$(dx/dy)^2 = \left(\tfrac{1}{2}y^3 - \tfrac{1}{2y^3}\right)^2$$

$$= \tfrac{1}{4}y^6 - \tfrac{1}{2} + \tfrac{1}{4y^6}$$

"one of those"

4. (10 points) Determine if the following improper integral is convergent or divergent. If convergent, evaluate.

$$\int_2^3 \frac{5x}{\sqrt{x^2-4}}\,dx \qquad \lim_{t\to 2^+} \int_t^3 (x^2-4)^{-1/2}\,5x\,dx =$$

$$\lim_{t\to 2^+} \frac{5}{2}\cdot\frac{1}{1/2}(x^2-4)^{1/2}\Big]_t^3 =$$

$$\lim_{t\to 2^+} 5\sqrt{x^2-4}\Big]_t^3 = \lim_{t\to 2^+}\left[5\sqrt{9-4} - 5\sqrt{t^2-4}\right] =$$

$$\left[5\sqrt{5} - 5\sqrt{0}\right] = 5\sqrt{5}$$ so Convergent

5. (10 points) Determine whether the series is convergent or divergent and if it is convergent **find the sum**.

a) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2+2n}$

$$\frac{A}{n} + \frac{B}{n+2} = \frac{1}{n(n+2)}$$

$A(n+2) + Bn = 1$ telescoping

$n=0$ $\quad A = 1/2$

$n=-2$ $\quad B = -1/2$

$$\sum_{n=1}^{\infty}\left(\frac{1/2}{n} - \frac{1/2}{n+2}\right)$$

$$\frac{\frac{1}{2}}{1} + \frac{1/2}{2} + \cancel{\frac{1/2}{3}} + \cancel{\frac{1/2}{4}} + \cancel{\frac{1/2}{5}} + \cdots$$

$$-\cancel{\frac{1/2}{3}} - \cancel{\frac{1/2}{4}} - \cancel{\frac{1/2}{5}}$$

$\frac{1}{2} + \frac{1}{4} = 3/4$ I found the sum of the series so the series is convergent

b) $\displaystyle\sum_{n=1}^{\infty} \frac{5^{n+1}}{4^{2n-2}}$ $\qquad \frac{5^2(5^{n-1})}{16^{n-1}}$

$$\sum_{n=1}^{\infty} \frac{25(5^{n-1})}{16(16^{n-1})} \qquad \frac{25}{16}\left(\frac{5}{16}\right)^{n-1}$$

$|r| = \frac{5}{16} < 1$ so convergent

and the sum $= \frac{a}{1-r}$

$$S = \frac{25/16}{1-5/16} = \frac{25}{16-5}$$

$25/11$

$b_n = \frac{1}{\sqrt{n^2+3n+1}}$

Compare to $\sum_{n=1}^{\infty} \frac{1}{n}$ a divergent p-series $p = 1 \le 1$

6. (10 points) Test the following series for absolute convergence, conditional convergence or divergence. State the test you are using and show all your work.

3

a) $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n^2+3n+1}}$

Limit Comparison Test,

$\lim_{n\to\infty} \left(\frac{1}{n} \cdot \frac{\sqrt{n^2+3n+1}}{1} \right) = 1 > 0$ and finite

so Both are divergent

Now Alternating Series Test

i) $\frac{1}{\sqrt{(n+1)^2+3(n+1)+1}} \le \frac{1}{\sqrt{n^2+3n+1}}$

ii) $\lim_{n\to\infty} \frac{1}{\sqrt{n^2+3n+1}} = 0$

so $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n^2+3n+1}}$ is Conditionally Convergent

3

b) $\sum_{n=1}^{\infty} \frac{n}{n^2+3^n}$ Compare to

$\sum_{n=1}^{\infty} \frac{1}{3^n}$ which is a convergent geometric series $r = 1/3 < 1$

Direct Comparison Test

$\frac{1}{n^2+3^n} \le \frac{1}{3^n}$

$3^n \le n^2+3^n$

$0 \le n^2$ true for all $n \ge 1$ so

$\sum_{n=1}^{\infty} \frac{1}{n^2+3^n}$ is Absolutely Convergent

4

c) $\sum_{n=1}^{\infty} \frac{(3 \cdot 7 \cdot 11 \cdot 15 \cdots (4n-1))}{5^n\, n!}$

Ratio Test

$\lim_{n\to\infty} \left| \frac{3 \cdot 7 \cdot 11 \cdots (4n-1)(4n+3)}{5 \cdot 5^n\, (n+1)\, n!} \cdot \frac{5^n\, n!}{3 \cdot 7 \cdot 11 \cdots (4n-1)} \right|$

$\lim_{n\to\infty} \left| \frac{4n+3}{5(n+1)} \right| = 4/5 < 1$ so Absolutely Convergent

6 7.

7. (~~10~~ points) The given curve is rotated about the y-axis. Find the area of the resulting surface. $S = 2\pi \int radius\ (arclength)$ $\quad y^{2/3} + x^{2/3} = 1, \quad 0 \le y \le 1$

$$S = 2\pi \int_0^1 (1-y^{2/3})^{3/2} \sqrt{} \qquad x = (1-y^{2/3})^{3/2}$$

2

$$S = 2\pi \int_0^1 (1-y^{2/3})^{3/2} \sqrt{1 + \frac{1-y^{2/3}}{y^{2/3}}}\, dy$$

$$dx/dy = \frac{3}{2}(1-y^{2/3})^{1/2}\left(-\frac{2}{3}y^{-1/3}\right) = -\frac{\sqrt{1-y^{2/3}}}{y^{1/3}}$$

$$S = 2\pi \int_0^1 (1-y^{2/3})^{3/2} \sqrt{\frac{y^{2/3} + 1 - y^{2/3}}{y^{2/3}}}\, dy \qquad (dx/dy)^2 = \frac{1-y^{2/3}}{y^{2/3}}$$

$$2\pi \int_0^1 (1-y^{2/3})^{3/2} \sqrt{\frac{1}{y^{2/3}}}\, dy$$

$$2\pi \int_0^1 (1-y^{2/3})^{3/2} \cdot \frac{1}{y^{1/3}}\, dy$$

$$u = 1 - y^{2/3}$$
$$du = -\tfrac{2}{3} y^{-1/3}\, dy$$
$$-\tfrac{3}{2} y^{1/3}\, du = dy$$

$$2\pi \int -\frac{3}{2} u^{3/2}\, du$$

$$2\pi \cdot -\frac{3}{2} \cdot \frac{1}{5/2} u^{5/2} = 2\pi\left(\frac{-3}{2}\right)\left(\frac{2}{5}\right)(1-y^{2/3})^{5/2}\Big]_0^1$$

$$2\pi\left(-\frac{3}{5}\right)\left[(1-1)^{5/2} - (1-0)^{5/2}\right]$$

$$-\frac{6\pi}{5}[-1] = 6\pi/5$$

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MATH 152 **EXAM IIA** NAME Solutions

Mrs. Bonny Tighe 50 points

7.7-11.7 TA_____ Wed 11/3/2021

PART II

1. (10 points) Complete the following tests

GEOMETRIC SERIES $\sum_{n=0}^{\infty} a_n (r)^n$ is Convergent if $|r| < 1$

THE COMPARISON TEST Suppose $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are series with positive terms.

i) If $\sum_{n=1}^{\infty} b_n$ is convergent and $a_n \leq b_n$ for all n, then $\sum_{n=1}^{\infty} a_n$ is also convergent

ii) If $\sum_{n=1}^{\infty} b_n$ is divergent and $a_n \geq b_n$ for all n, then $\sum_{n=1}^{\infty} a_n$ is also divergent

THE LIMIT COMPARISON TEST

Suppose that $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are series with positive terms . If $\lim_{n\to\infty} \frac{a_n}{b_n} = c$ where c is positive + finite, then either both series converge or both diverge.

Name the test: Test for Divergence If $\lim_{n\to\infty} a_n \neq 0$, then $\sum_{n=1}^{\infty} a_n$ is divergent

THE INTEGRAL TEST Suppose f is a positive, continuous and decreasing function on $[1, \infty)$ and let $a_n = f(n)$ then the series $\sum_{n=1}^{\infty} a_n$ is convergent if and only if $\int_a^{\infty} f(n)\,dn$ is convergent.

P-SERIES $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is divergent if $p \leq 1$

RATIO TEST

If $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = L$

$L < 1$ then $\sum a_n$ is absolutely convergent

$L > 1$ then $\sum a_n$ is divergent

$L = 1$ then the test is inconclusive

ROOT TEST

If $\lim_{n\to\infty} \sqrt[n]{|a_n|} = L$ and

$L < 1$ then $\sum a_n$ is absolutely convergent

$L > 1$ then $\sum a_n$ is divergent

$L = 1$ then the root test is inconclusive

THE ALTERNATING SERIES TEST

If i) if $b_{n+1} \leq b_n$ for all n and ii) $\lim_{n\to\infty} b_n = 0$ Then $\sum_{n=1}^{\infty} (-1)^{n-1} b_n$ is Convergent

REMAINDER ESTIMATE FOR THE INTEGRAL TEST

Suppose $a_n = f(n)$ where f is a continuous, positive, decreasing function for $x \geq n$, and $\sum a_n$ is convergent . If $R_n = S - S_n$, then $\int_{n+1}^{\infty} f(n)\,dn \leq R_n \leq \int_n^{\infty} f(n)\,dn$

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2. (~~10~~ points). Use the **Integral test** to determine if the series is divergent or convergent.

$$\sum_{n=1}^{\infty} \frac{e^{1/n}}{n^2}$$

$u = \frac{1}{n}$ $du = -\frac{1}{n^2}dn$

$$\lim_{t\to\infty} \int_1^t e^{1/n}\left(\frac{1}{n^2}\right)dn = \lim_{t\to\infty} -e^{1/n}\Big]_1^t$$

$$\lim_{t\to\infty} \left[-e^{1/t} - (-e^{1/1})\right] = (-1+e)$$ which is finite

$\sum_{n=1}^{\infty} \frac{e^{1/n}}{n^2}$ is also convergent

8/10

3. (10 points) Find the length of the arc of the curve $y = \frac{x^3}{6} + \frac{1}{2x}$, $\frac{1}{2} \le x \le 1$

$y = \frac{1}{6}x^3 + \frac{1}{2}x^{-1}$

$dy/dx = \frac{1}{2}x^2 - \frac{1}{2}x^{-2}$

$= \frac{1}{2}x^2 - \frac{1}{2x^2}$

$(dy/dx)^2 = \frac{1}{4}x^4 - \frac{1}{2} + \frac{1}{4x^4}$

$$L = \int_{1/2}^{1} \sqrt{1 + (dy/dx)^2}\, dx$$

$$= \int_{1/2}^{1} \sqrt{\frac{1}{4}x^4 + \frac{1}{2} + \frac{1}{4x^4}}\, dx$$

"one of those"

$$\int_{1/2}^{1} \sqrt{\left(\frac{1}{2}x^2 + \frac{1}{2x^2}\right)^2}\, dx = \int_{1/2}^{1} \left(\frac{1}{2}x^2 + \frac{1}{2}x^{-2}\right) dx$$

$$\frac{1}{2}\cdot\frac{1}{3}x^3 + \frac{1}{2}\cdot\frac{1}{-1}x^{-1}\Big]_{1/2}^{1} = \frac{1}{6}x^3 - \frac{1}{2x}\Big]_{1/2}^{1} =$$

$$\left(\frac{1}{6} - \frac{1}{2}\right) - \left(\frac{1}{6}\left(\frac{1}{8}\right) - \frac{1}{1}\right) = \frac{1}{6} - \frac{1}{2} - \frac{1}{48} + 1$$

$$-\frac{2}{6} - \frac{1}{48} + 6/6$$

$$\left(\frac{8}{8}\right) 4/6 - \frac{1}{48}$$

$$\frac{32 - 1}{48} = \frac{31}{48}$$

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4. ~~(10~~ 5 points) Determine if the following improper integral is convergent or divergent. If convergent, evaluate.

discontinuous $\int_1^3 \frac{x}{\sqrt{x^2-1}}\,dx$

$\lim_{t\to 1^+} \int_t^3 (x^2-1)^{-1/2}\, x\,dx =$ $\quad u = x^2-1$, $du = 2x\,dx$

$\lim_{t\to 1^+} \frac{1}{2}\cdot\frac{1}{1/2}(x^2-1)^{1/2}\Big]_t^3 = \lim_{t\to 1^+} \sqrt{x^2-1}\Big]_t^3 =$

$\lim_{t\to 1^+}\left(\sqrt{9-1} - \sqrt{t^2-1}\right) = \sqrt{8} - 0 = \sqrt{8}$ a finite limit

So $\int_1^3 \frac{x}{\sqrt{x^2-1}}\,dx$ is convergent.

5. (10 points) Determine whether the series is convergent or divergent and if it is convergent **find the sum.**

a) $\sum_{n=1}^{\infty} \frac{1}{n^2+3n}$

$\frac{A}{n} + \frac{B}{n+3} = \frac{1}{n(n+3)}$

$A(n+3) + Bn = 1$

$n=0 \quad A = 1/3$

$n=-3 \quad B = -1/3$

$\sum_{n=1}^{\infty}\left(\frac{1/3}{n} - \frac{1/3}{n+3}\right) =$

$\frac{1/3}{1} + \frac{1/3}{2} + \frac{1/3}{3} + \frac{1/3}{4} + \frac{1/3}{5} + \frac{1/3}{6}$

$-\frac{1/3}{4} \quad -\frac{1/3}{5} \quad -\frac{1/3}{6}$

telescopic

The sum of the series is

$\frac{1}{3}+\frac{1}{6}+\frac{1}{9}$ because I found the sum, the series $\sum_{n=1}^{\infty}\frac{1}{n^2+3n}$ is convergent

b) $\sum_{n=1}^{\infty} \frac{(-4)^n}{e^{2n+2}}$ $\quad \frac{(-4)(-4)^{n-1}}{(e^2)^{n+1}}$

$\sum_{n=1}^{\infty} \frac{(-4)(-4)^{n-1}}{e^4(e^2)^{n-1}}$ $\quad r = \left|\frac{-4}{e^2}\right| < 1$

So Convergent by the Geometric series test

$\text{Sum} = \frac{a}{1-r}$

$S = \frac{-4/e^4}{1-(-4/e^2)} =$

$\frac{-4}{e^4+4e^2}$

11/18

10

6. (10 points) Test the following series for absolute convergence, conditional convergence or divergence. State the test you are using and show all your work.

$b_n = \frac{1}{2n+3}$

3 a) $\sum_{n=1}^{\infty} \frac{n+1}{\sqrt{n^5+3n^2+1}}$

Compare to $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$ a convergent p-series $p = 3/2 > 1$

Limit Comparison Test

$\lim_{n\to\infty} \left(\frac{n+1}{\sqrt{n^5+3n^2+1}} \cdot \frac{n^{3/2}}{1} \right) =$

$\lim_{n\to\infty} \left(\frac{n^{7/2}+n^{3/2}}{\sqrt{n^5+3n^2+1}} \right) = 1 > 0$ and finite

$\therefore \sum_{n=1}^{\infty} \frac{n+1}{\sqrt{n^5+3n^2+1}}$ is Absolutely Convergent

4 b) $\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+3}$

Compare to $\sum_{n=1}^{\infty} \frac{1}{n}$ a divergent p-series $p = 1 \le 1$

– Limit Comparison Test:

$\lim_{n\to\infty} \left(\frac{1}{n} \cdot \frac{2n+3}{1} \right) = 2 > 0$ and finite so both are divergent.

– next use the Alternating Series Test

i) $\frac{1}{2(n+1)+3} \le \frac{1}{2n+3}$

ii) $\lim_{n\to\infty} \left(\frac{1}{2n+3} \right) = 0$

$\therefore \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+3}$ is Conditionally Convergent

3 c) $\sum_{n=1}^{\infty} \frac{4 \cdot 7 \cdot 10 \cdots (3n+1)}{4^n n!}$

Ratio Test

$\lim_{n\to\infty} \left| \frac{(4 \cdot 7 \cdot 10 \cdots (3n+1)(3n+4))}{4 \cdot 4^n (n+1) n!} \cdot \frac{4^n n!}{(4 \cdot 7 \cdot 10 \cdots (3n+1))} \right|$

$\lim_{n\to\infty} \left| \frac{3n+4}{4(n+1)} \right| = 3/4 < 1$ so absolutely convergent

5 $\qquad\qquad$ $r=y$

7. (~~10~~ points) The given curve is rotated about the x-axis. Find the area of the resulting surface. $S = 2\pi\int_a^b radius\,arclength$ $\qquad y^{2/3}+x^{2/3}=1,\ 0\le x\le 1$

dx

$$S = 2\pi\int_0^1 y\sqrt{1+(dy/dx)^2}\,dx$$

$$y = (1-x^{2/3})^{3/2}$$

$$dy/dx = \frac{3}{2}(1-x^{2/3})^{1/2}\left(-\frac{2}{3}x^{-1/3}\right)$$

$$= \frac{-\sqrt{1-x^{2/3}}}{x^{1/3}}$$

$$(dy/dx)^2 = \frac{1-x^{2/3}}{x^{2/3}}$$

$$S = 2\pi\int_0^1 (1-x^{2/3})^{3/2}\sqrt{1+\frac{1-x^{2/3}}{x^{2/3}}}\,dx$$

$$S = 2\pi\int_0^1 (1-x^{2/3})^{3/2}\sqrt{\frac{x^{2/3}+1-x^{2/3}}{x^{2/3}}}\,dx$$

$$2\pi\int_0^1 (1-x^{2/3})^{3/2}\sqrt{\frac{1}{x^{2/3}}}\,dx = 2\pi\int_0^1 (1-x^{2/3})^{3/2}\left(\frac{1}{x^{1/3}}\right)dx$$

$u = 1-x^{2/3}$

$du = -2/3\,x^{-1/3}\,dx$

$= dx$

$-\frac{3}{2}x^{1/3}\,du$

$$2\pi\int -\frac{3}{2}u^{3/2}\,du$$

$$2\pi\ -3/2\cdot\frac{1}{5/2}u^{5/2}\Big]$$

$$2\pi\left(-\frac{3}{2}\cdot\frac{2}{5}\right)(1-x^{2/3})^{5/2}\Big]_0^1$$

$$2\pi\left(-\frac{3}{5}\right)\left[0^{5/2}-1^{5/2}\right] = -\frac{6}{5}\pi[-1] = 6\pi/5$$

MATH 152
Mrs. Bonny Tighe

EXAM IIA

50 points
7.7-11.7
PART I

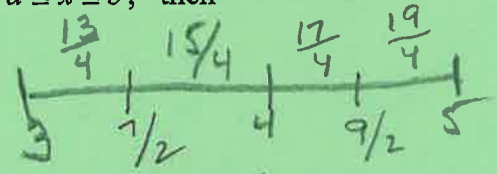
NAME Solutions

TA_____ Wed 11/3/2021

1. (10 points) Use (a) the Midpoint Rule (b) the Trapezoidal Rule and (c) Simpson's Rule to approximate the given integral with the specified value of n. Find the error bounds for the Midpoint rule. Suppose $|f''(x)| \le K$ *for* $a \le x \le b$, then

$\Delta x = 1/2$

$$|E_M| \le \frac{K(b-a)^3}{24n^2} \qquad \int_3^5 \ln x\, dx, \quad n = 4$$



$$M_4 = \frac{1}{2}\left[\ln(13/4) + \ln(15/4) + \ln(17/4) + \ln(19/4)\right]$$

$$T_4 = \frac{1}{4}\left[\ln 3 + 2\ln(\tfrac{7}{2}) + 2\ln 4 + 2\ln(9/2) + \ln 5\right]$$

$$S_4 = \frac{1}{6}\left[\ln 3 + 4\ln(\tfrac{7}{2}) + 2\ln 4 + 4\ln(9/2) + \ln 5\right]$$

$f(x) = \ln x$

$f'(x) = \frac{1}{x}$

$f''(x) = \left|-\frac{1}{x^2}\right| \le K$

on $[3, 5]$

largest at 3

$K = \frac{1}{9}$

$$|E_M| \le \frac{\frac{1}{9}(5-3)^3}{24(16)}$$

$$|E_M| \le \frac{\cancel{8}}{9(24)\cancel{(16)}_{2}} = \frac{1}{18(24)}$$

How large should we take n in order to guarantee that the Midpoint approximation for

$\int_3^5 \ln x\, dx, \quad n = 4$ is accurate to within 0.001?

$$\frac{\frac{1}{9}(8)}{24n^2} \le \frac{1}{10^3}$$

$$\frac{1}{9(3)n^2} \le \frac{1}{10^3}$$

$$10^3 \le 9(24)n^2$$

$$\frac{10^3}{27} \le n^2$$

$$n \ge \sqrt{\frac{10^3}{27}}$$

The next largest integer

2. (15 points) Multiple Choice: Circle the correct letter

1 2 3 4

a) Find a sequence for $1, \frac{1\cdot 5}{1}, \frac{1\cdot 5\cdot 9}{1\cdot 2}, \frac{1\cdot 5\cdot 9\cdot 13}{1\cdot 2\cdot 3}, \ldots$ is

a) $a_n = \frac{4n-3}{n!}$

b) $a_n = \frac{(1\cdot 5\cdot 9\cdots(4n-3))}{(n-1)!}$

c) $a_n = \frac{(1\cdot 5\cdot 9\cdots(4n-3))}{n!}$

d) $a_n = \frac{4n-1}{(n-1)!}$

e) None of the above

b) Which of the following sequences is not convergent?

a) $a_n = \frac{5^n+2}{3^n}$

b) $a_n = \frac{4n-1}{2n+1}$

c) $a_n = \frac{3}{n!}$

d) $a_n = \ln(n+2) - \ln(n+3)$

e) They are all convergent

c) $\sum_{n=1}^{\infty} \frac{(-3)^{n+1}}{2^{2n}}$ equals

a) $\frac{-12}{7}$

b) $\frac{9}{7}$

c) 12

d) $\frac{-9}{7}$

e) None of the above

d) Which test would you use to determine the absolute convergence of the series $\sum_{n=0}^{\infty} \frac{(-1)^n}{e^n}$

a) Limit Comparison Test
b) Root Test
c) Integral Test
d) Ratio Test
e) None of the above

All of the above

e) Which test would you use to determine the convergence of the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n^3+4n+2}}$

a) Alternating Series Test
b) Integral Test
c) Limit Comparison Test
d) Ratio Test
e) All of the above tests

3. (5 points) Circle True or False

Test for divergence

a) **True** **False** The first test you try when determining the convergence of a series is the ratio test.

b) **True** **False** Given $a_n = n^2 + 3n + 1$, then $a_{n+1} = n^2 + 5n + 5$.

$(n+1)^2 + 3(n+1) + 1$
$n^2 + 2n + 1 + 3n + 3 + 1$

c) **True** **False** To determine the convergence of $\sum_{n=0}^{\infty} \frac{n-2}{n^2+n^3+1}$ I could use the Integral Test.

d) **True** **False** If $\lim_{n\to\infty} a_n = 0$, then the sequence $\{a_n\}$ is convergent.

e) **True** **False** If $\lim_{n\to\infty} a_n = 0$, then the series $\sum_{n=0}^{\infty} a_n$ is convergent.

may or may not be

4. (10 points) Circle True or False

True **False** a) The formula for the Trapezoid rule for approximation of a definite integral with four subdivisions is $\Delta x\left[f(x_0)+2f(x_1)+2f(x_2)+2f(x_3)+f(x_4)\right]$.

True **False** b) We would compare $\int_1^\infty \frac{x}{\sqrt{x^6+2x-3}}dx$ to $\int_1^\infty \frac{1}{x^2}dx$ to determine convergence.

True **False** c) The Surface Area for the function $y=e^{2x}$ on the interval $0\le x\le 1$ is

$$2\pi\int_0^1 e^{2x}\sqrt{1+4e^{2x^2}}\,dx$$

$dy/dx = e^{2x}(2)$ $(dy/dx)^2 = 4e^{4x}$

True **False** d) $\sum_{n=2}^{\infty}\frac{(-1)^n n}{\ln(n)}$ is conditionally convergent . Divergent

True **False** e) The arc length for the function $y=\tan x$ on the interval $0\le y\le 1$ is

$$\int_0^1 \sqrt{1+\left(\sec^4 x\right)}dx$$

dy

5. (15 points) Multiple Choice – (3 points each) Circle the correct letter

a) When using the comparison test for the improper integral $\int_2^\infty \frac{x}{\sqrt{x^5-2x^3-3}}dx$ the inequality will be

$\frac{1}{x^{3/2}}$

a) $\frac{1}{\sqrt{x^5}}\le\frac{x}{\sqrt{x^5-2x^3-3}}$

b) $\frac{1}{\sqrt{x^5}}\ge\frac{x}{\sqrt{x^5-2x^3-3}}$

c) $\frac{1}{\sqrt{x^3}}\le\frac{x}{\sqrt{x^5-2x^3-3}}$

d) $\frac{1}{\sqrt{x^3}}\ge\frac{x}{\sqrt{x^5-2x^3-3}}$

e) None of the above

b) Represent $0.\overline{746}$ as a series

a) $\sum_{n=0}^{\infty} \left(\frac{746}{1000}\right)^n$

b) $\sum_{n=0}^{\infty} \frac{746}{1000}\left(\frac{1}{10}\right)^n$ c) $\sum_{n=0}^{\infty} 746\left(\frac{1}{1000}\right)^n$

d) $\sum_{n=0}^{\infty} \frac{746}{1000}\left(\frac{1}{1000}\right)^n$

e) None of the above

c) The surface area for the function $x = e^y$ when rotated about the y-axis on the interval $2 \le y \le 3$ is:

$r = x$

a) $2\pi \int_2^3 y\sqrt{1+e^{2y}}\,dy$

b) $2\pi \int_2^3 e^y \sqrt{1+e^{2y}}\,dy$

c) $2\pi \int_2^3 x\sqrt{1+\left(\frac{1}{x^2}\right)}\,dx$

d) $2\pi \int_2^3 \ln x\sqrt{1+\left(\frac{1}{x^2}\right)}\,dx$

e) None of the above

d) The K for the error bound for the midpoint formula for approximating $\int_1^4 \frac{1}{\sqrt{x}}\,dx$ with 4 subdivisions is:

a) $K = -\frac{3}{4}$

b) $K = \frac{3}{128}$

c) $K = \frac{3}{4}$

d) $K = \frac{1}{2}$

e) None of the above

e) Which of the following is **not** an improper integral?

a) $\int_{-2}^{5} \frac{1}{\sqrt{x+2}} dx$

b) $\int_{1}^{3} \frac{x}{x^2+3x-4} dx$

c) $\int_{1}^{3} \frac{x}{x^2-3x+4} dx$

d) $\int_{0}^{1} \frac{1}{\ln x} dx$

e) They are all improper integrals